CS Notes (Groups)

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Outline

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Introduction

Creating Groups in Python

Permutations with SymPy

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Python

We will use Python in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups. We will primarily interact with Python using the Spyder IDE.

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Python Fundamentals

The fundamental objects we need from Python are:

- lists: `[obj1, obj2, ...]`
- functions: `def func(x): ...`

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Creating Groups in Python

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A Simple Group in Python

Suppose we want to model the coins group of the previous class. Then we will need to specify the *objects* and the *actions* of the group. We can use lists to hold the objects...

```
# group from 1.1 has as objects arrangements of a penny and a
↪  dime
arrangement1 = ['penny', 'nickel']
arrangement2 = ['nickel', 'penny']
objects = [arrangement1, arrangement2]
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A Simple Group in Python

...and functions to represent the actions:

```python
# our action takes arrangement1 and turns it into arrangement2, and
# vice-versa
def group_action(arrangement):
    if arrangement[0] == 'penny':
        return arrangement2
    else:
        return arrangement1

# check that this does what we want it to
for object in objects:
    print(group_action(object))
```

```
['nickel', 'penny']
['penny', 'nickel']
```


A Simple Group in Python
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Group Actions in Python

Example (Combining group actions using Python)

Find the result of performing the group action twice in a row using the code above.

Example (Words group again)

Implement the "three letter words" group in Python. Be careful to specify the objects and the actions!

Group Actions in Python
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There should be six objects and six actions that act on the objects!

Recall that a symmetry group of a geometric figure is the set of actions that preserves the shape of that figure.

Example (Symmetry group of equilateral triangle)

Implement the symmetry group of the equilateral triangle in Python.

Symmetry Groups in Python

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Example (Symmetry group of equilateral triangle)

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This is more complicated than previous examples. Try working out examples by hand, then implementing in Python with a `for` loop.

Permutations with SymPy

Permutations with SymPy

Permutations in Python

- The SymPy library lets us handle permutations in cycle notation relatively easily.
- First, we need to import useful commands as follows:

```
from sympy.combinatorics import Permutation
from sympy.combinatorics import PermutationGroup
from sympy.combinatorics import SymmetricGroup
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Computations from S_3

- We can now create permutations as follows:

```
R = Permutation(1, 2, 3)
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Example (Rotation followed by flip)

Let R denote clockwise rotation of the equilateral triangle by 120° and let F denote the reflection that leaves vertex 1 unchanged. Using SymPy, find $R * F$ and $F * R$.

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More Computations

Example (Klein Four Group)

The symmetry group of a non-square rectangle is called the *Klein Four Group*. Create the elements of this group in Python using SymPy as above. Does the order of "multiplication" matter in this group?

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Example (Cayley table)

Find the Cayley table of the symmetry group of the square

Even More Computations
Example (Cayley table)
Find the Cayley table of the symmetry group of the square

Break into four groups; each group tackles two rows of the table